# Project Report

## Visualization Tool for Electric Vehicle Charge and Range Analysis

Date: 09 July 2026

# Table of Contents

# 1. Introduction

## 1.1 Project Overview

Electric vehicles (EVs) are vehicles powered by an electric motor that draws electricity from a battery, which can be charged from an external source, rather than relying on an internal combustion engine. While EVs are not a new concept, they have received significantly more attention in recent years as advances in battery technology and analytics have driven increased market share.

Modern EVs combine electrical storage and propulsion systems with electronic sensors, controls, and actuators, integrated closely with software, secure data transfer, and data analysis, to form a comprehensive transportation solution. The common thread running through all of these advances is data analytics — and this project, the "Visualization Tool for Electric Vehicle Charge and Range Analysis," is built around that idea.

The project brings together charging station data, vehicle sensor data, and EV model specifications into a single Tableau-based analytics platform, serving three distinct personas: a city energy planner analyzing charging demand, an EV fleet manager monitoring battery performance and range, and a prospective EV buyer comparing models before purchase.

## 1.2 Purpose

The purpose of this project is to give every EV stakeholder a single, trustworthy view of charging, battery, and vehicle data — replacing scattered spreadsheets and manual reports with interactive dashboards that turn raw EV data into clear, actionable insight. Specifically, the tool aims to:

- Help city energy planners understand charging demand patterns and plan new infrastructure.
- Help fleet managers monitor battery health and driving range to reduce downtime and cost.
- Help prospective EV buyers compare models on price, range, efficiency, and performance so they can make an informed, cost-effective purchase decision.

# 2. Ideation Phase

## 2.1 Problem Statement

A well-articulated customer problem statement allows the team to find the ideal solution for the challenges customers face, and helps the team empathize with how customers perceive the product. The following problem statements were defined using the “I am / I'm trying to / But / Because / Which makes me feel” template:

| PS | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | Ravi, an EV Buyer | Identify the most efficient EV brands | Comparing multiple models and brands is time-consuming. | I want to make an informed and cost-effective purchase decision. | Confused while selecting an EV |
| PS-2 | Hasini, a City Energy Planner | Analyze EV charging stations across different regions. | Charging station information is scattered across multiple sources. | I need accurate insights to plan future charging stations. | Uncertain about infrastructure planning. |

## 2.2 Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviors and attitudes. Creating the map helped the team consider the problem from the user's perspective, along with their goals and challenges. The persona “Aarav” below represents the combined viewpoint of an EV buyer, city planner, and fleet manager.

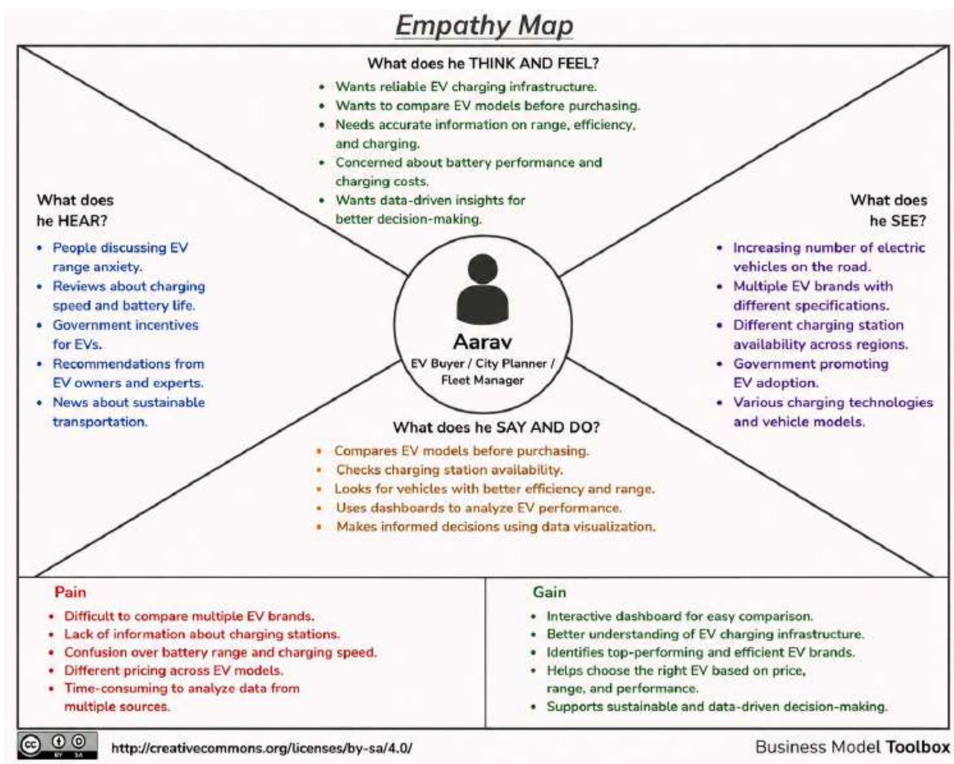


*Figure 2.1: Empathy map for the combined EV Buyer / City Planner / Fleet Manager persona*

The following expanded empathy map breaks the same four quadrants — Think & Feel, Hear, See, and Say & Do — into individual observations, along with Pain and Gain points gathered during the exercise.

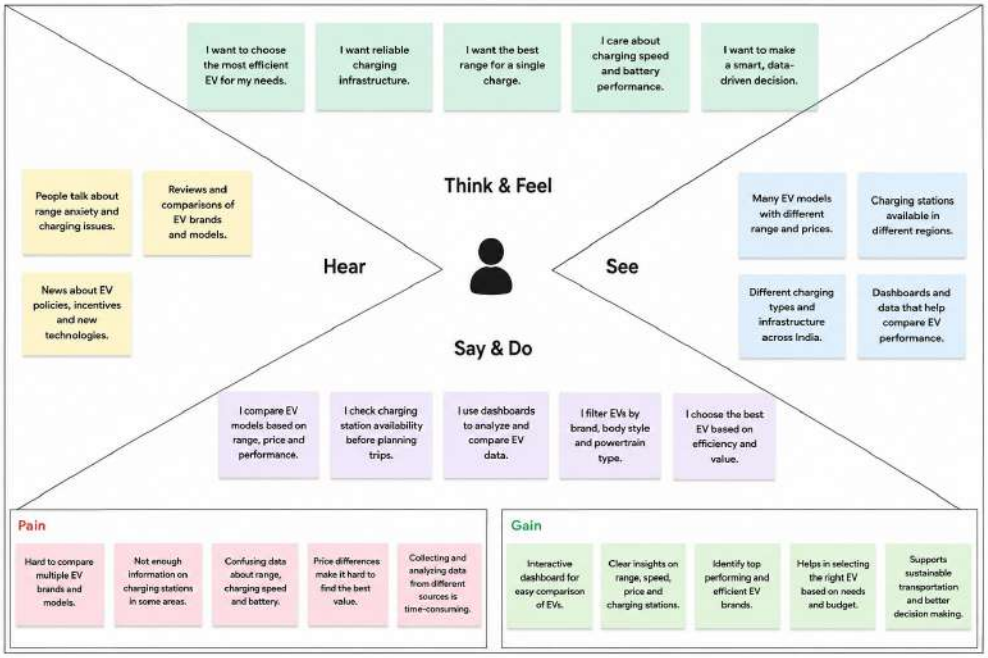
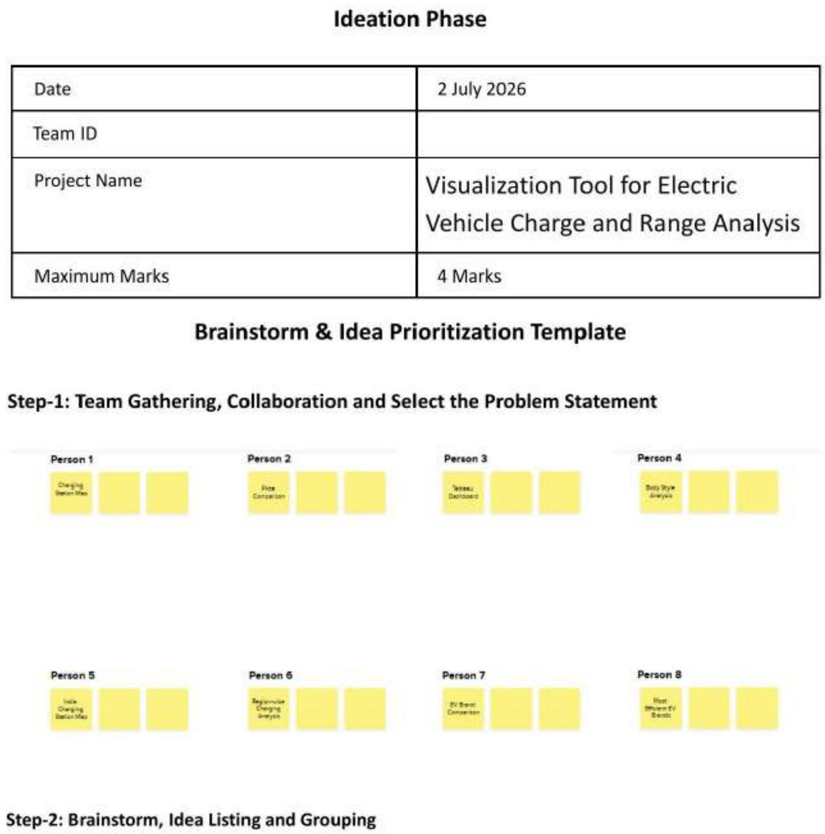



*Figure 2.2: Detailed empathy map with individual observations*

## 2.3 Brainstorming

The brainstorming exercise followed three steps: team gathering and problem selection, idea listing and grouping, and idea prioritization.

**Step 1: Team Gathering, Collaboration and Selecting the Problem Statement**



**Ideation Phase**

| | |
|---|---|
| Date | 2 July 2026 |
| Team ID | |
| Project Name | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 4 Marks |

**Brainstorm & Idea Prioritization Template**

**Step-1: Team Gathering, Collaboration and Select the Problem Statement**



**Step-2: Brainstorm, Idea Listing and Grouping**

*Figure 2.3: Each team member contributes an initial idea*

**Step 2: Brainstorm, Idea Listing and Grouping**


*Figure 2.4: Ideas listed and grouped into a shared board*

### Step 3: Idea Prioritization

Each idea was plotted on an Importance vs. Feasibility matrix to decide what to build first. Ideas in the top-right quadrant (high importance, high feasibility) — Charging Station Map, EV Brand Comparison, Price Comparison, Battery Range Analysis, and Efficient EV Brands — were prioritized for the first release.

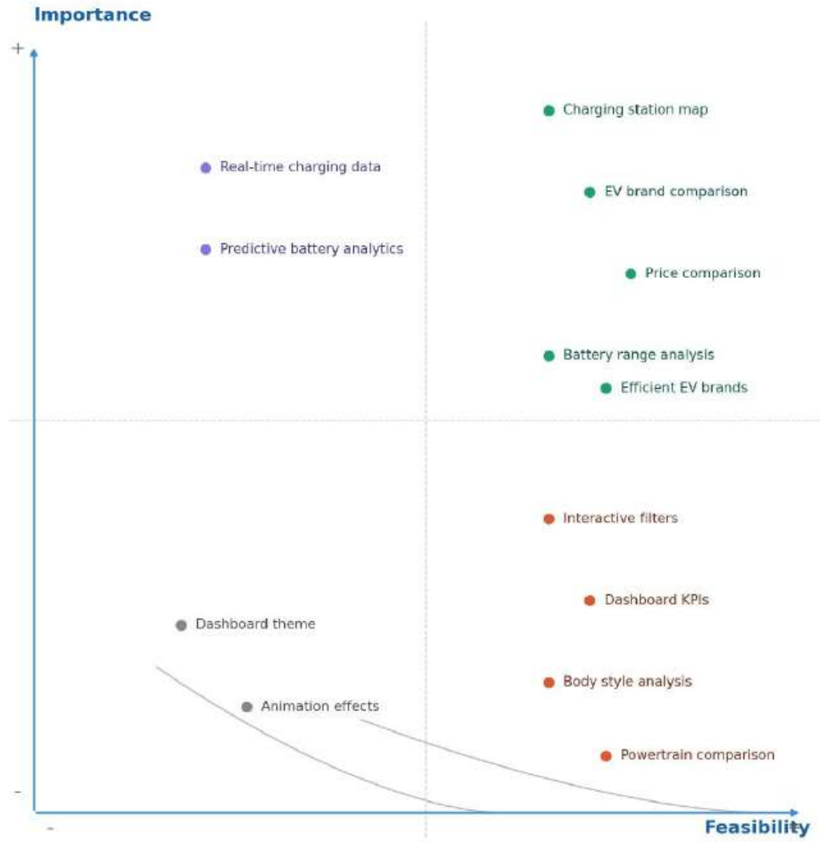



*Figure 2.5: Idea prioritization matrix (Importance vs. Feasibility)*

# 3. Requirement Analysis

## 3.1 Customer Journey Map

The customer journey map traces how each persona — city energy planner, EV fleet manager, and prospective EV buyer — discovers, explores, and acts on EV charging, battery, and model data through the visualization dashboard. It is organized into five phases: Entice, Enter, Engage, Exit, and Extend.

### Entice

*How does someone initially become aware of this tool?*

| Step | Interactions | Goals & Motivations | Positive Moments | Negative Moments |
|---|---|---|---|---|
| **Hears About the Dashboard** | Word of mouth, city notice, or online search for "EV comparison India" | Help me find a reliable way to understand EV options | Excited to hear one tool covers charging, range, and pricing together | Unsure if the tool covers their city or preferred brands |

*Areas of opportunity:*

| | |
|---|---|
| **Hears About the Dashboard** | Could we run city/dealer partnerships to increase awareness? |

### Enter

*What do people experience as they begin the process?*

| Step | Interactions | Goals & Motivations | Positive Moments | Negative Moments |
|---|---|---|---|---|
| **Visits Website / App** | Website / Tableau Public / Tableau Server link | Help me quickly see what this tool offers | Clear, role-based options (planner, fleet manager, buyer) feel welcoming | Non-technical users may feel overwhelmed by dashboard jargon |
| **Browses Homepage** | Landing page listing the Charging, Battery & Range, and EV Comparison dashboards | Help me understand which dashboard is relevant to me | Filtering feels fast and responsive | Too many filter options can feel overwhelming at first |
| **Selects a Use Case** | Dashboard navigation tabs / menu | Help me get to the data that matters to me | The heatmap makes peak charging hours immediately obvious | Some users may not know their own driving conditions to compare against |

*Areas of opportunity:*

| | |
|---|---|
| **Visits Website / App** | Add a short intro banner explaining the three dashboards |
| **Browses Homepage** | Add simple role-based entry points ("I'm a planner", "I'm a buyer") |
| **Selects a Use Case** | Offer preset filter combinations (e.g., "Budget EVs", "Family SUVs") |

### Engage

*In the core moments in the process, what happens?*

| Step | Interactions | Goals & Motivations | Positive Moments | Negative Moments |
|---|---|---|---|---|
| **Applies Filters** | Filter panel (brand, region, price, body style, powertrain type) | Help me narrow results down to only what's relevant | Comparing predicted vs. actual range builds confidence in the data | Users may skip the story if it isn't clearly signposted |
| **Views Charging Dashboard** | Charging heatmap, region/type bar chart, station map | Help me understand where and when demand is highest | The heatmap makes peak charging hours immediately obvious | Non-technical users may feel overwhelmed by dashboard jargon |
| **Views Battery & Range Dashboard** | Predicted vs. actual range chart, efficiency-by-brand chart | Help me trust the real-world range, not just the spec sheet | Comparing predicted vs. actual range builds confidence in the data | Some users may not know their own driving conditions to compare against |
| **Views EV Model Comparison** | Price, top speed, and body-style comparison charts | Help me compare models side-by-side without visiting many websites | Having price, speed, and body style together saves a lot of time | Users may skip the story if it isn't clearly signposted |
| **Follows Guided Story** | Tableau Story walking through charging, pricing, and brand data | Help me see the big picture, not just individual charts | The story format makes the data feel like a narrative, not a spreadsheet | Exported PDFs lose some interactivity |

*Areas of opportunity:*

| | |
|---|---|
| **Applies Filters** | Add a simple driving-style selector to personalize range estimates |
| **Views Charging** | Add a "Start here" prompt pointing new users to the story first |

| | |
|---|---|
| **Dashboard** | |
| **Views Battery & Range Dashboard** | Add a simple driving-style selector to personalize range estimates |
| **Views EV Model Comparison** | Offer preset filter combinations (e.g., “Budget EVs”, “Family SUVs”) |
| **Follows Guided Story** | Add a “Start here” prompt pointing new users to the story first |

## Exit

*What do people experience as the process finishes?*

| Step | Interactions | Goals & Motivations | Positive Moments | Negative Moments |
|---|---|---|---|---|
| **Exports Report** | Export / Download PDF button | Help me save what I found so I can compare later or share it | Having all the data in one place makes the decision feel well-informed | Exported PDFs lose some interactivity |
| **Makes a Decision** | Final comparison view / saved filters | Help me feel confident I've picked the right EV, station plan, or fleet action | Easy to share a link rather than re-explaining everything | No notification when new, relevant data is added |

*Areas of opportunity:*

| | |
|---|---|
| **Exports Report** | Offer an option to export a personalized summary, not just raw charts |
| **Makes a Decision** | Add optional email alerts for new models or stations matching saved filters |

## Extend

*What happens after the experience is over?*

| Step | Interactions | Goals & Motivations | Positive Moments | Negative Moments |
|---|---|---|---|---|
| **Shares the Dashboard** | Shareable link or exported PDF sent by email | Help me get input from family, colleagues, or my team | Easy to share a link rather than re-explaining everything | No notification when new, relevant data is added |
| **Returns for Updates** | Revisits dashboard as new models or stations are added | Help me stay current as the market changes | Appreciate that data refreshes without needing a new tool | No notification when new, relevant data is added |
| **Gives Feedback** | Feedback form / rating prompt | Help me help improve the tool for the next person | Appreciate that data refreshes without needing a new tool | Feedback forms can feel like a chore |

*Areas of opportunity:*

| | |
|---|---|
| **Shares the Dashboard** | Add optional email alerts for new models or stations matching saved filters |
| **Returns for Updates** | Add optional email alerts for new models or stations matching saved filters |
| **Gives Feedback** | Make feedback a quick 1-tap rating with an optional comment |

## 3.2 Solution Requirement

### Functional Requirements:

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) |
|---|---|---|
| FR-1 | Data Ingestion & Integration | Import EV charging station data into MySQL<br>Import vehicle specification and battery sensor data<br>Connect the MySQL data source to Tableau |
| FR-2 | Data Preprocessing & Cleaning | Remove duplicate records and handle missing (NULL) values<br>Split columns (e.g., CheapElectricCars) and rename fields (e.g., Brands)<br>Change field data types (e.g., Address to Geographic Role) for map visualizations |
| FR-3 | Dashboard & Visualization | Charging pattern & usage analysis dashboard<br>Battery performance vs. driving range dashboard<br>Comparative EV model efficiency dashboard |
| FR-4 | Filtering & Interactivity | Filter by brand, region, price, and body style<br>Filter by powertrain type and vehicle name<br>Export dashboards and stories as PDF reports |

### Non-functional Requirements:

| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-1 | Usability | Dashboards should be intuitive and easy to navigate for non-technical users, including city planners, fleet managers, and everyday EV consumers, with minimal training required. |
| NFR-2 | Security | Access to charging, vehicle, and location data should be role-based and protected, ensuring only authorized users can view or export sensitive information. |
| NFR-3 | Reliability | The system should consistently deliver accurate, up-to-date data and visualizations with minimal errors or downtime. |
| NFR-4 | Performance | Dashboards should load and refresh within a few seconds, even when processing large datasets across multiple regions and vehicle brands. |
| NFR-5 | Availability | The platform should be available 24/7 so city planners, fleet managers, and consumers can access insights whenever needed. |
| NFR-6 | Scalability | The system should scale to support additional cities, EV brands, charging stations, and users without requiring a redesign of the core architecture. |

## 3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.


*Figure 3.1: High-level data flow*

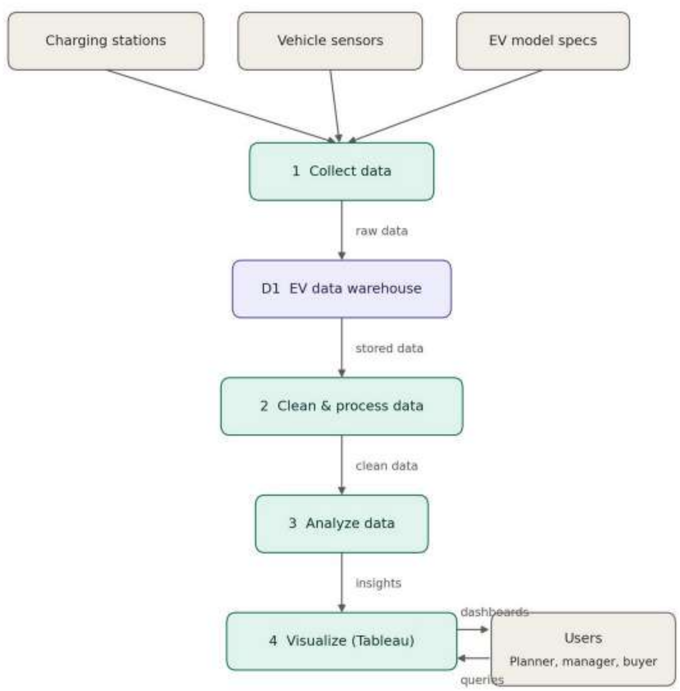



*Figure 3.2: Detailed data flow diagram*

**User Stories:**

User stories for the EV analytics and visualization dashboard, covering the city energy planner, fleet manager, and consumer personas.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priori ty | Rele ase |
|---|---|---|---|---|---|---|
| City energy planner | Charging pattern analysis | USN-1 | As a city energy planner, I can view dashboards of charging time, duration, and power consumed across the city so I can understand EV charging demand. | Dashboard shows charging trends by time of day and location | High | Sprin t-1 |

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| City energy planner | Peak demand visualization | USN-2 | As a city energy planner, I can view heatmaps of neighborhoods with high EV charging demand so I can plan new charging infrastructure. | Heatmap highlights high-demand areas by neighborhood | High | Sprint-1 |
| Fleet manager | Battery performance monitoring | USN-3 | As a fleet manager, I can track state of charge, depth of discharge, and efficiency for each vehicle so I can monitor battery health. | Dashboard displays battery health metrics per vehicle | High | Sprint-1 |
| Fleet manager | Range prediction | USN-4 | As a fleet manager, I can compare predicted versus actual driving range so I can spot underperforming vehicles. | Dashboard shows predicted vs actual range side by side | Medium | Sprint-2 |
| Fleet manager | Maintenance alerts | USN-5 | As a fleet manager, I can receive an alert when a vehicle's battery is likely to need replacement so I can reduce downtime. | Alert is triggered when battery health falls below a set threshold | Medium | Sprint-2 |
| Consumer (EV buyer) | Model comparison | USN-6 | As a consumer, I can compare different EV models on price, range, and efficiency so I can choose the best vehicle for my needs. | Dashboard shows a side-by-side comparison of multiple EV models | High | Sprint-1 |
| Consumer (EV buyer) | Cost & emissions estimate | USN-7 | As a consumer, I can view cost per kilometer and estimated long-term savings for each EV model so I can make an informed financial decision. | Dashboard displays cost and emissions estimates per model | Medium | Sprint-2 |
| Consumer (EV buyer) | Personalized filtering | USN-8 | As a consumer, I can filter EV model comparisons by driving style and region so I get personalized recommendations. | Filters update the comparison dashboard by driving style and region | Low | Sprint-2 |

## 3.4 Technology Stack

The diagram below shows the architecture of the EV Analytics & Visualization Dashboard, from data sources through cloud processing to the Tableau dashboards used by planners, fleet managers, and consumers.

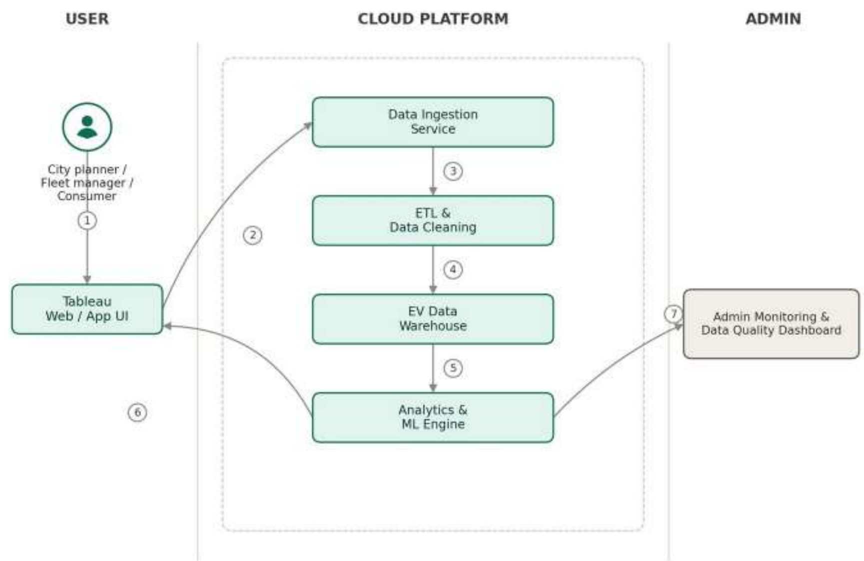



*Figure 3.3: Technology stack architecture*

### Table-1: Components & Technologies:

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | Interactive dashboards used by city planners, fleet managers, and consumers to explore EV data | Tableau Desktop / Tableau Server, HTML, CSS, JavaScript |
| 2. | Application Logic-1 | Ingests raw data from charging stations, vehicle sensors, and EV model specs | Python (Pandas, Requests), REST APIs |
| 3. | Application Logic-2 | Cleans, transforms, and aggregates raw data (ETL) | Python, SQL, Tableau Prep Builder |
| 4. | Application Logic-3 | Runs analytics and predictive models for battery health and range prediction | Python (scikit-learn), R |
| 5. | Database | Structured storage of charging, battery, and vehicle data | MySQL / PostgreSQL |
| 6. | Cloud Database | Cloud-hosted EV data warehouse for analytics at scale | AWS RDS, Google BigQuery, or Snowflake |
| 7. | File Storage | Storage for raw sensor logs and exported reports | AWS S3 / Google Cloud Storage |
| 8. | External API-1 | Charging station location and availability data | Open Charge Map API |
| 9. | External API-2 | EV model specifications and pricing data | Manufacturer / EV database API (e.g., NREL) |
| 10. | Machine Learning Model | Predicts battery degradation and driving range | Regression / Random Forest model (scikit-learn) |
| 11. | Infrastructure (Server / Cloud) | Application deployment. Local Server Configuration: N/A. Cloud Server Configuration: Cloud VM for ETL jobs + Tableau Server | AWS EC2, Tableau Server on Cloud |

### Table-2: Application Characteristics:

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Data processing and modeling built on open-source Python libraries | Pandas, scikit-learn, Tableau Prep Builder |
| 2. | Security Implementations | Data encrypted at rest and in transit; access to dashboards and data restricted by role | AES-256 encryption, TLS, Tableau Server role-based access control (RBAC), OAuth2 |
| 3. | Scalable Architecture | Layered pipeline (ingestion, storage, analytics, visualization) hosted on the cloud, allowing each layer to scale independently as vehicles and users grow | Cloud data warehouse, containerized ETL jobs |
| 4. | Availability | Cloud database replication and Tableau Server clustering keep dashboards accessible with minimal downtime | Load-balanced Tableau Server, cloud DB replicas |

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 5. | Performance | Tableau extracts and cached queries reduce load time; scheduled ETL jobs keep the warehouse fresh without slowing dashboards | Tableau data extracts, indexed SQL queries |

# 4. Project Design

## 4.1 Problem – Solution Fit

The Problem-Solution Fit simply means that the team has found a real problem with the customer, and that the proposed solution actually solves that problem. It helps identify behavioral patterns and recognize what would work and why.

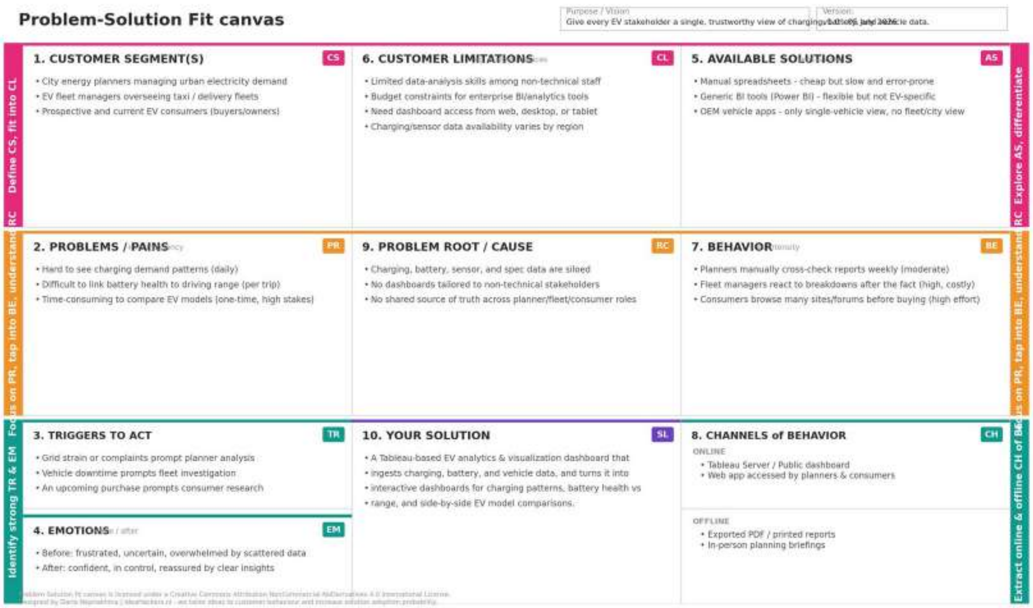



*Figure 4.1: Problem-Solution Fit canvas*

## 4.2 Proposed Solution

| S.No. | Parameter | Description |
|---|---|---|
| 1. | Problem Statement (Problem to be solved) | City planners, fleet managers, and everyday EV buyers all struggle to make sense of EV data. Planners can't easily see charging demand patterns, fleet managers can't tell how battery health is affecting real driving range, and consumers have no easy way to compare EV models on more than price. The data exists, but it's scattered and not visualized in one place. |
| 2. | Idea / Solution description | A Tableau-based EV Analytics & Visualization Dashboard that ingests charging station, vehicle sensor, and EV model data, cleans and stores it in a central warehouse, and presents it through three tailored dashboards: charging pattern & usage analysis, battery performance vs. driving range, and comparative EV model efficiency. |
| 3. | Novelty / Uniqueness | Brings three very different stakeholders (city planner, fleet manager, consumer) onto one platform with role-specific views, directly correlates battery health metrics with real-world driving range instead of just showing raw specs, and lets consumers filter model comparisons by their own driving style and region for a personalized recommendation. |
| 4. | Social Impact / Customer Satisfaction | Helps cities plan charging infrastructure and manage grid load more sustainably, reduces fleet downtime and maintenance costs through proactive battery insights, and gives consumers the confidence to choose the right EV — all of which support faster, smoother EV adoption and lower emissions. |
| 5. | Business Model (Revenue Model) | SaaS subscriptions for city/government agencies and fleet operators (tiered by number of vehicles or seats); a freemium consumer-facing comparison tool with premium features (personalized reports, alerts) behind a paywall; and data-partnership or licensing revenue from charging networks and EV manufacturers. |
| 6. | Scalability of the Solution | The cloud-hosted data pipeline and warehouse scale horizontally as more vehicles, charging stations, and users come online. The modular architecture allows new data sources (additional EV brands, new cities) and new dashboard types to be added without redesigning the core system. |

## 4.3 Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

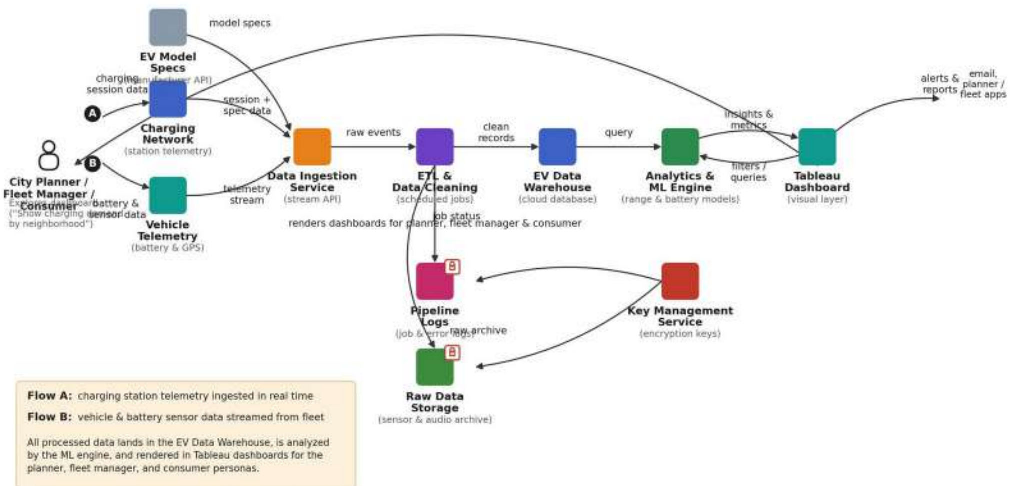



*Figure 4.2: Architecture and data flow of the EV Analytics & Visualization Dashboard*

# 5. Project Planning & Scheduling

## 5.1 Project Planning

**Product Backlog & Sprint Schedule:**

| Sprint | Functional Requirement (Epic) | USN | User Story / Task | Points | Priority | Team |
|---|---|---|---|---|---|---|
| Sprint-1 | Data Integration & Preprocessing | USN-1 | As a data engineer, I can import EV charging, vehicle, and battery datasets into MySQL and connect them to Tableau so all data is available for analysis. | 5 | High | |
| Sprint-1 | Data Integration & Preprocessing | USN-2 | As a data engineer, I can remove duplicate records and handle missing/NULL values so the dataset is clean and reliable. | 3 | High | |
| Sprint-1 | Data Integration & Preprocessing | USN-3 | As a data engineer, I can split the CheapElectricCars column into three columns and rename Split 3 to Brands so brand data is usable. | 5 | High | |
| Sprint-1 | Data Integration & Preprocessing | USN-4 | As a data engineer, I can change the Address field's data type to Geographic Role so map-based visualizations work correctly. | 3 | Medium | |
| Sprint-1 | Data Integration & Preprocessing | USN-5 | As a developer, I can validate and correct data types across text, numeric, and categorical fields so calculations are accurate. | 4 | Medium | |
| Sprint-2 | Charging Pattern & Usage Dashboard | USN-6 | As a city energy planner, I can view a heatmap of EV charging demand by time of day and duration so I can identify peak charging hours. | 5 | High | |
| Sprint-2 | Charging Pattern & Usage Dashboard | USN-7 | As a city energy planner, I can filter charging stations by region and connector type so I can plan new infrastructure. | 5 | High | |
| Sprint-2 | Charging Pattern & Usage Dashboard | USN-8 | As a city energy planner, I can view charging stations plotted on a map of India so I can see their geographic distribution. | 4 | Medium | |
| Sprint-2 | Charging Pattern & Usage Dashboard | USN-9 | As a user, I can view a bar chart of charging stations grouped by region and type. | 3 | Medium | |
| Sprint-2 | Charging Pattern & Usage Dashboard | USN-10 | As a user, I can export the charging dashboard as a PDF report. | 3 | Low | |
| Sprint-3 | Battery Performance & Range Dashboard | USN-11 | As a fleet manager, I can compare predicted vs. actual driving range for each vehicle so I can spot underperforming vehicles. | 5 | High | |
| Sprint-3 | Battery Performance & Range Dashboard | USN-12 | As a fleet manager, I can view efficiency (Wh/Km) by brand so I can identify top-performing EV brands. | 5 | High | |
| Sprint-3 | Battery Performance & Range Dashboard | USN-13 | As a fleet manager, I can view top speed by brand so I can compare vehicle performance. | 4 | Medium | |
| Sprint-3 | Battery Performance & Range Dashboard | USN-14 | As a fleet manager, I can filter vehicles by brand, body style, and powertrain type. | 3 | Medium | |
| Sprint-3 | Battery Performance & Range Dashboard | USN-15 | As a user, I can view calculated fields such as BodyStyle_Count and Car_Brands_India for aggregated analysis. | 3 | Low | |
| Sprint-4 | EV Model Comparison & Story | USN-16 | As a consumer, I can compare price ranges of different EV models available in India so I can choose within my budget. | 5 | High | |
| Sprint-4 | EV Model Comparison & Story | USN-17 | As a consumer, I can view different EV car models side-by-side so I can compare specifications. | 5 | High | |
| Sprint-4 | EV Model Comparison & Story | USN-18 | As a consumer, I can filter models by brand, price, and region for a personalized comparison. | 4 | Medium | |
| Sprint-4 | EV Model Comparison & Story | USN-19 | As a user, I can view a guided Tableau Story that walks through charging stations, pricing, and brand comparisons. | 3 | Medium | |
| Sprint-4 | EV Model Comparison & Story | USN-20 | As a user, I can view the final dashboards published to Tableau Public/Server for sharing. | 3 | Low | |

**Project Tracker, Velocity & Burndown Chart:**

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Sprint Release Date (Actual) |
|---|---|---|---|---|---|
| Sprint-1 | 20 | 6 Days | 08 Jul 2026 | 13 Jul 2026 | |
| Sprint-2 | 20 | 6 Days | 14 Jul 2026 | 19 Jul 2026 | |
| Sprint-3 | 20 | 6 Days | 20 Jul 2026 | 25 Jul 2026 | |
| Sprint-4 | 20 | 6 Days | 26 Jul 2026 | 31 Jul 2026 | |

Velocity: With a 10-day sprint duration and a team velocity of 20 story points per sprint, the team's average velocity (AV) per iteration unit is AV = sprint duration / velocity = 20 / 10 = 2 story points per day.

Burndown Chart: A burndown chart is a graphical representation of work remaining versus time, commonly used in agile methodologies such as Scrum, to track progress across each sprint against the planned story points.

# 6. Functional and Performance Testing

## 6.1 Performance Testing

Performance testing verifies that the dashboards remain fast, accurate, and reliable under realistic data volumes and usage patterns. The table below lists the planned test cases; the Result and Actual columns should be filled in with measured values once testing is carried out on the deployed dashboards.

| S.No | Test Case | Description | Expected Result | Actual Result |
|---|---|---|---|---|
| 1. | Dashboard load time | Measure time to load the Charging, Battery & Range, and EV Comparison dashboards | Loads within 3 seconds | |
| 2. | Filter response time | Measure time for a dashboard to update after applying a filter (brand, region, price, etc.) | Updates within 1 second | |
| 3. | Concurrent users | Test dashboard responsiveness with multiple simultaneous users on Tableau Server | No noticeable slowdown up to 50 concurrent users | |
| 4. | Data refresh / ETL job duration | Measure time for scheduled ETL jobs to refresh the EV data warehouse | Completes within the scheduled maintenance window | |
| 5. | Export performance | Measure time to export a dashboard or story as a PDF report | Completes within 5 seconds | |
| 6. | Large dataset handling | Test dashboard performance as charging/vehicle records scale into the hundreds of thousands | Maintains load and filter times within target thresholds | |

# 7. Results

## 7.1 Output Screenshots

*This section should include screenshots of the completed Tableau dashboards — the Charging Pattern & Usage dashboard, the Battery Performance vs. Range dashboard, and the Comparative EV Model Efficiency dashboard — along with the guided Tableau Story. Insert screenshots from the published dashboard here before submitting the final report.*

| |
|---|
| *[ Insert dashboard screenshot here ]* |

## 8. Advantages & Disadvantages

**Advantages:**

- Brings charging, battery, and EV model data into a single, unified view for three very different stakeholders.
- Turns raw sensor and station data into clear, actionable dashboards without requiring technical expertise.
- Correlates battery health with real-world driving range, giving more trustworthy insight than spec sheets alone.
- Cloud-hosted, scalable architecture that can grow to support new cities, brands, and users.
- Supports sustainable EV adoption by helping planners, fleets, and consumers make better-informed decisions.

**Disadvantages:**

- Depends on the quality, completeness, and availability of third-party data (charging networks, manufacturer specs).
- Tableau licensing and cloud infrastructure introduce ongoing costs.
- Predictive models for battery health and range require continuous retraining as new data arrives.
- Initial ETL and data-cleaning setup can be time-consuming, especially when integrating multiple data sources.
- Dashboard performance may degrade with very large datasets unless extracts and queries are carefully optimized.

## 9. Conclusion

The Visualization Tool for Electric Vehicle Charge and Range Analysis brings together charging station data, vehicle sensor data, and EV model specifications into a single Tableau-based platform. By tailoring dashboards to three distinct personas — a city energy planner, an EV fleet manager, and a prospective EV buyer — the project turns scattered, hard-to-interpret EV data into clear, role-specific insight.

Through the ideation, requirement analysis, and design phases, the project identified real customer problems, mapped them to a concrete technical architecture, and translated them into a prioritized backlog of user stories delivered across four sprints. The result is a scalable, cloud-hosted analytics pipeline that supports smarter charging infrastructure planning, healthier EV fleets, and more confident EV purchase decisions.

## 10. Future Scope

- Extend the dashboard to a dedicated mobile app for on-the-go access by fleet managers and consumers.
- Incorporate real-time streaming data from charging stations and vehicle telematics, rather than periodic batch refreshes.
- Add anomaly-detection models to flag unusual battery degradation or charging behavior automatically.
- Expand data coverage to additional EV brands, regions, and countries beyond the initial dataset.
- Introduce multilingual support to make the dashboards accessible to a wider range of users.
- Add gamification or personalized recommendations to increase consumer engagement with the comparison tool.

## 11. Appendix

**Dataset Link:**

https://drive.google.com/drive/folders/1FypJgZvFsZIHRiZqBKZLqjJj8Q3LZ0k-?usp=sharing

**GitHub & Project Demo Link:**

**GitHub:** https://github.com/dolaurmila/Visualization-Tool-for-Electric-Vehicle-Charge-and-Range-Analysis
**Demo Video:** https://drive.google.com/file/d/1RJBYEnAwOi4nLi9QWnntse60BQjl3vhk/view?usp=sharing